

Product Release Notes — ChatAssist v2.3

New Features, Improvements, Bug Fixes, and Known Issues

AI & Data Science Team | Release Date: March 15, 2026

Build: 2.3.0-rc4 | Branch: release/2.3 | Classification: Internal

1. Release Overview

ChatAssist v2.3 is a major release focused on multilingual support, improved retrieval accuracy, and Slack integration. This release includes 47 commits across 23 PRs, with contributions from 8 team members. The release has been validated in UAT with 50 internal beta testers over 2 weeks, achieving a 4.2/5.0 satisfaction score.

Metric	v2.2 (Previous)	v2.3 (Current)	Change
Retrieval Accuracy (benchmark)	87.3%	91.2%	+3.9pp
TTFT (Time to First Token)	1.8s	1.3s	-27.8%
Full Response Time (median)	6.1s	5.2s	-14.8%
Documents Indexed	1,847	2,312	+25.2%
Supported Languages	1 (English)	3 (English, Vietnamese, mixed)	+2 languages
Daily Active Users (beta)	28	42	+50%
User Satisfaction (beta)	3.8/5.0	4.2/5.0	+10.5%

2. New Features

Feature	Description	PR #	Author
Multilingual Support	ChatAssist now supports Vietnamese and mixed English-Vietnamese queries. Uses multilingual-e5-large embedding model for cross-language retrieval. Vietnamese queries can find English documents and vice versa. Language detection is automatic.	#247	Le Thi Hong
Slack Integration	Users can query ChatAssist directly from Slack via the /ask command or by mentioning @ChatAssist in any channel. Responses include source citations and a 'View Full Answer' link to the web UI. Supports thread-based follow-up questions.	#251	Nguyen Duc Anh
Source Attribution UI	Each answer now shows numbered source citations [1][2][3] with expandable preview cards. Clicking a citation opens the original document at the relevant section. Confidence scores are displayed per source.	#238	Do Thanh Tam
Admin Analytics Dashboard	New admin page showing: daily query volume, popular topics, unanswered questions, user feedback trends, document coverage gaps. Exportable to CSV.	#244	Pham Van Khanh
Feedback Loop	Users can rate answers (thumbs up/down) and provide text feedback. Negative feedback triggers automatic review queue for content team. Weekly digest sent to document owners.	#242	Dr. Tran Minh Khoa
Conversation Memory	ChatAssist maintains context across multi-turn conversations (up to 10 turns). Users can reference previous answers ('like you mentioned earlier'). Session persists for 30 minutes of inactivity.	#249	Le Thi Hong

3. Improvements

Area	Improvement	Impact	PR #
Retrieval	Upgraded to hybrid search (vector + BM25 with RRF fusion). Alpha parameter tuned to 0.7 for optimal balance.	+3.9pp retrieval accuracy	#235
Retrieval	Added cross-encoder reranker (bge-reranker-base) as second-stage ranking. Top-40 candidates reranked to top-6.	+2.1pp precision@5	#236
Chunking	Implemented semantic chunking based on sentence embedding similarity. Replaces fixed 500-token chunks with topic-aware boundaries.	Chunks are more coherent, +15% relevance	#240
Ingestion	Parallel document processing: 4 workers, batch embedding (32 docs). Progress bar in admin UI.	3x faster ingestion (100 pages/min)	#245
UI	Redesigned chat interface with message streaming, code syntax highlighting, and markdown rendering.	Improved readability and UX score	#243
Performance	Response caching for repeated queries (Redis, TTL 1 hour). Cache hit rate: ~22% in beta.	40% reduction in LLM API calls	#248
Security	Added per-user document access control. Users only see results from documents they have permission to access.	RBAC compliance for sensitive docs	#250

4. Bug Fixes

Bug ID	Description	Root Cause	Fix	Severity
BUG-1823	Table data not extracted from multi-page PDFs spanning 3+ pages	pdfplumber page cropping excluded table continuation rows	Extended table detection to merge cross-page table fragments	Medium
BUG-1847	Answer hallucination when no relevant documents found	LLM generated plausible but incorrect answers when retrieval returned low-confidence results	Added confidence threshold (0.3); below threshold returns 'I don't have enough information'	High
BUG-1856	Duplicate chunks created when re-uploading same document	Document deduplication checked filename only, not content hash	Added SHA-256 content hash comparison before ingestion	Medium
BUG-1861	Special characters in queries (e.g., C++, .NET) causing search failures	Query tokenizer stripped special characters before embedding	Added character preservation for known technical terms	Low
BUG-1872	Session timeout not properly clearing conversation context	Redis TTL not applied to conversation history keys	Fixed Redis key expiry to match session TTL (30 min)	Low
BUG-1879	Vietnamese diacritics not preserved in document titles after ingestion	UTF-8 encoding lost during PDF text extraction normalization	Added Unicode NFC normalization, preserving Vietnamese characters	Medium

5. Known Issues & Limitations

Issue	Description	Workaround	Planned Fix
KNOWN-001	Large tables (>50 rows) may be truncated in responses due to LLM context window limits	Ask about specific rows or filter criteria	v2.4 — table summarization
KNOWN-002	Scanned PDFs (image-based) are not yet supported; only text-based PDFs are ingested	Convert scanned PDFs to text using Adobe Acrobat OCR before upload	v2.5 — Tesseract OCR integration
KNOWN-003	Response streaming occasionally stutters on slow network connections (<5 Mbps)	Refresh the page if streaming stops	v2.4 — WebSocket reconnection
KNOWN-004	Excel (.xlsx) and PowerPoint (.pptx) files are parsed but table layout may be imperfect	Convert complex spreadsheets to PDF before upload	v2.5 — improved spreadsheet parser

6. Upgrade Instructions

This is a backward-compatible release. No database migration required. Existing document indexes are compatible with v2.3. To take advantage of the new multilingual embedding model, a full re-indexing is recommended (approximately 4 hours for the current 2,312-document corpus).

Step	Command	Duration	Notes
1. Pull latest image	<code>docker pull techviet/chatassist:2.3.0</code>	2 minutes	Verify SHA: sha256:a1b2c3...
2. Update config	<code>cp config.yaml.example config.yaml && edit</code>	5 minutes	New: MULTILINGUAL_ENABLED, SLACK_BOT_TOKEN
3. Run migrations	<code>python manage.py migrate</code>	30 seconds	Adds feedback, analytics tables
4. Restart services	<code>docker compose up -d</code>	1 minute	Health check: /api/v1/health
5. Re-index (optional)	<code>python scripts/reindex.py --model=multilingual-e5-large</code>	~4 hours	Run during off-hours; non-blocking